

RAMCO INSTITUTE OF TECHNOLOGY

**DEPARTMENT OF ELECTRONICS AND
COMMUNICATION ENGINEERING**

COURSE OUTCOMES

REGULATION 2017

COURSE OUTCOMES – BATCH 2017-2021	
I Year	
Course Code & Title : HS8151 Communicative English Semester & Year of study : I & 2017-18 Course Index : C101	
Couse Index	Course Outcomes
C101.1	Communicate clearly both in the written form and orally using appropriate vocabulary and comprehend written texts to make inferences.
C101.2	Speak persuasively in different social contexts and write biographical details and technical documents cohesively, coherently and flawlessly using appropriate words.
C101.3	Speak, read and write effectively for a variety of professional and social settings.
C101.4	Read descriptive, narrative, expository and interpretive texts and write using creative, critical, analytical and evaluative methods.
C101.5	Listen, comprehend and respond to different spoken and written discourses/excerpts in different accents and write different genres of texts adopting various writing strategies.
Course Code & Title : MA8151 Engineering Mathematics - I Semester & Year of study : I & 2017-18 Course Index : C102	
Couse Index	Course Outcomes
C102.1	Use both the limit definition and rules of differentiation to differentiate functions.
C102.2	Apply differentiation to solve maxima and minima problems.
C102.3	Evaluate integrals both by using Riemann sums and by using the Fundamental Theorem of Calculus, also evaluate integrals using techniques of integration, such as substitution, partial fractions and integration by parts, in addition to determine convergence/divergence of improper integrals and evaluate convergent improper integrals.
C102.4	Apply integration to compute multiple integrals, area, volume, integrals in polar coordinates, in addition to change of order and change of variables.
C102.5	Apply various techniques in solving differential equations.
Course Code & Title : PH8151 Engineering Physics Semester & Year of study : I & 2017-18 Course Index : C103	
Couse Index	Course Outcomes
C103.1	Analyze the elastic nature of materials and be able to choose the materials depending upon the modulus of elasticity for different applications.
C103.2	Illustrate the advantages of optical communication using LASER.

C103.3	Explain the conducting properties of solids, liquids, good thermal conductor and bad thermal conductors
C103.4	Apply the knowledge of quantum mechanics and classical mechanics in addressing the problems related to science and technology
C103.5	Describe the crystal structures, crystal defects and various crystal growth techniques.
Course Code & Title : CY8151 Engineering Chemistry Semester & Year of study : I & 2017-18 Course Index : C104	
Couse Index	Course Outcomes
C104.1	Describe the importance of water technology in the purification of water and its domestic and industrial applications.
C104.2	Illustrate the concept of absorption in surface chemistry and catalysis and its applications.
C104.3	Review use of the phase rule in identifying its application in metallurgy and manufacture of alloys.
C104.4	Compare the different types of industrial techniques of petroleum processing and the determination of caloric values and combustion parameters.
C104.5	Explain the fundamentals of different alternative source of energy, the generation process and batteries.
Course Code & Title : GE8151 Problem Solving and Python Programming Semester & Year of study : I & 2017-18 Course Index : C105	
Couse Index	Course Outcomes
C105.1	Develop algorithmic solutions to simple computational problems.
C105.2	Read, write and execute simple python programs.
C105.3	Apply control, looping structures and functions to solve problems.
C105.4	Represent compound data using python lists, tuples, and dictionaries.
C105.5	Read and Write data from/to files in python programs.
Course Code & Title : GE8152 Engineering Graphics Semester & Year of study : I & 2017-18 Course Index : C106	
Couse Index	Course Outcomes
C106.1	Familiarize with the fundamentals and standards of Engineering graphics
C106.2	Perform freehand sketching of basic geometrical constructions and multiple views of objects.

C106.3	Project orthographic projections of lines and plane surfaces.
C106.4	Draw projections and section of solids and development of surfaces.
C106.5	Visualize and to project isometric and perspective sections of simple solids.
Course Code & Title : GE8161 Problem Solving and Python Programming Laboratory Semester & Year of study : I & 2017-18 Course Index : C107	
Couse Index	Course Outcomes
C107.1	Write, test, and debug simple Python programs.
C107.2	Implement Python programs with conditionals and loops.
C107.3	Develop Python programs step-wise by defining functions and calling them.
C107.4	Demonstrate the use Python lists, tuples, and dictionaries for representing compound data.
C107.5	Illustrate the concepts of read and write data from/to files in Python.
Course Code & Title : BS8161 Physics and Chemistry Laboratory Semester & Year of study : I & 2017-18 Course Index : C108	
Couse Index	Course Outcomes
C108.1	Test materials by using their knowledge of applied physics principles in optics and properties of matter.
C108.2	Perform the quantitative chemical analysis of chloride, dissolved oxygen, hardness, alkalinity and copper ions by titration methods.
C108.3	Demonstrate basic concepts in the determination of acids, sodium, potassium and iron by the instrumental methods of analysis.
Course Code & Title : HS8251 Technical English Semester & Year of study : II & 2017-18 Course Index : C109	
Couse Index	Course Outcomes
C109.1	Read technical texts and write area specific texts effortlessly.
C109.2	Listen and comprehend lectures and talks in their areas of specialization and write effectively for a variety of professional and social settings.
C109.3	Speak and write appropriately and effectively in varied formal and informal contexts.
C109.4	Write effectively and persuasively and produce different types of writing such as letters, minutes, reports and winning job applications.

C109.5	Communicate clearly using technical vocabulary in their professional correspondences.
Course Code & Title : MA8251 Engineering Mathematics - II Semester & Year of study : II & 2017-18 Course Index : C110	
Couse Index	Course Outcomes
C110.1	Compute the Eigenvalues and eigenvectors, diagonalization of a matrix, Symmetric matrices, Positive definite matrices and similar matrices.
C110.2	Find Gradient, divergence and curl of a vector point function and related identities, Evaluation of line, surface and volume integrals using Gauss, Stokes and Green's theorems and their verification.
C110.3	Solve problems on Analytic functions and conformal mapping.
C110.4	Evaluate complex integrals.
C110.5	Find Laplace transform and inverse transform of simple functions, properties, various related theorems and application to differential equations with constant coefficients.
Course Code & Title : PH8253 Physics for Electronics Engineering Semester & Year of study : II & 2017-18 Course Index : C111	
Couse Index	Course Outcomes
C111.1	Explain the properties of conducting materials using classical and quantum concepts.
C111.2	Apply the fundamental knowledge about the semiconductors and able to differentiate different types of semiconductors.
C111.3	Explain the properties of Magnetic, Dielectric materials and devices for modern day to day applications.
C111.4	Explain the properties and applications of Optical materials and devices.
C111.5	Apply the knowledge about the nano-electronic materials and devices for various applications.
Course Code & Title : BE8254 Basic Electrical and Instrumentation Engineering Semester & Year of study : II & 2017-18 Course Index : C112	
Couse Index	Course Outcomes
C112.1	Explain the operation of three phase electrical circuits and power system.
C112.2	Determine the regulation and efficiency of transformers.
C112.3	Describe the characteristics of DC Generator and Motor.
C112.4	Analyze the performance of AC and DC machines.

C112.5	Apply the concepts of measurements and instruments for real time applications.
Course Code & Title : EC8251 Circuit Analysis Semester & Year of study : II & 2017-18 Course Index : C113	
Couse Index	Course Outcomes
C113.1	Determine current and voltage in circuits using Ohm's Law, Kirchhoff's laws, mesh current method, node voltage method and network topology.
C113.2	Apply the Network theorems to the analysis of AC and DC circuits.
C113.3	Calculate the response of the series and parallel resonance circuits, coupled circuits and tuned circuits.
C113.4	Solve first and second order AC and DC circuits for steady-state and transient response in the time domain using Laplace transforms.
C113.5	Understand the concept of two port network, its various parameters and symmetrical
Course Code & Title : EC8252 Electronic Devices Semester & Year of study : II & 2017-18 Course Index : C114	
Couse Index	Course Outcomes
C114.1	Explain the structure, operations and characteristics of PN Junction diode.
C114.2	Describe the basic geometry, operation and various configuration of Bipolar Junction Transistor.
C114.3	Analyze the operation of various Field Effect Transistors.
C114.4	Describe the operations of Special Semiconductor Devices.
C114.5	Explain the basic concepts of Power and Display devices.
Course Code & Title : EC8261 Circuits and Devices Laboratory Semester & Year of study : II & 2017-18 Course Index : C115	
Couse Index	Course Outcomes
C115.1	Analyze the characteristics of basic electronic devices.
C115.2	Determine the transient response of RL and RC circuits.
C115.3	Perform Kirchhoff's Current Law and Kirchhoff's Voltage Law.

C115.4	Verify Thevinin, Norton, Superposition, Maximum Power Transfer and Reciprocity Theorems.
C115.5	Determine the Resonant frequency of RLC circuits.
Course Code & Title : GE8261 Engineering Practices Laboratory Semester & Year of study : II & 2017-18 Course Index : C116	
Couse Index	Course Outcomes
C116.1	Construct carpentry components and pipe connections including plumbing works.
C116.2	Use welding equipment's to join the structures
C116.3	Illustrate the basic machining operations
C116.4	Construct the models using sheet metal works
C116.5	Describe centrifugal pump, Air conditioner, operations of smithy, foundry and fittings.
C116.6	Construct the basic Electrical and Electronics circuits.
C116.7	Examine the different types of electronic circuits and components.
C116.8	Explain the electrical safety rules, grounding, general house wiring.
C116.9	Perform soldering in various electronic circuits.
C116.10	Illustrate the basic operation of domestic electrical appliances.
II Year	
Course Code & Title : MA8352 Linear Algebra and Partial Differential Equations Semester & Year of study : III & 2018-19 Course Index : C201	
Couse Index	Course Outcomes
C201.1	Explain the fundamental concepts of advanced algebra and their role in modern mathematics and applied contexts, to demonstrate accurate and efficient use of advanced algebraic techniques and to demonstrate their mastery by solving non - trivial problems related to the concepts and by proving simple theorems about the statements proven by the text on the topic Vector Spaces.
C201.2	Explain the fundamental concepts of advanced algebra and their role in modern mathematics and applied contexts, to demonstrate accurate and efficient use of advanced algebraic techniques and to demonstrate their mastery by solving non -

	trivial problems related to the concepts and by proving simple theorems about the statements proven by the text on the topic Linear Transformation and Diagonalization.
C201.3	Explain the fundamental concepts of advanced algebra and their role in modern mathematics and applied contexts, to demonstrate accurate and efficient use of advanced algebraic techniques and to demonstrate their mastery by solving non - trivial problems related to the concepts and by proving simple theorems about the statements proven by the text on the topic Inner Product Spaces.
C201.4	Solve various types of partial differential equations.
C201.5	Solve engineering problems using Fourier series.
Course Code & Title : EC8393 Fundamentals of Data Structures In C Semester & Year of study : III & 2018-19 Course Index : C202	
Couse Index	Course Outcomes
C202.1	Develop the programs in C using basic constructs.
C202.2	Develop the programs in C using function, pointers, structures and unions.
C202.3	Suggest and Implement appropriate linear data structure operations for any given data set in C.
C202.4	Suggest and Implement appropriate non-linear data structure operations for a given application in C.
C202.5	Appropriately choose the sorting algorithms and also apply hashing concepts for a given problem.
Course Code & Title : EC8351 Electronic Circuits-I Semester & Year of study : III & 2018-19 Course Index : C203	
Couse Index	Course Outcomes
C203.1	Design the various biasing circuits of BJT, JFET and MOSFET.
C203.2	Analyze the small signal equivalent and design BJT amplifier circuits.
C203.3	Analyze the small signal equivalent and design JFET and MOSFET amplifier circuits.
C203.4	Plot the frequency response of all amplifiers.
C203.5	Design the regulated power supply, troubleshoot and analyze the faults in power supplies.

Course Code & Title : EC8352 Signals and Systems Semester & Year of study : III & 2018-19 Course Index : C204	
Couse Index	Course Outcomes
C204.1	Analyze the various properties of signals and systems.
C204.2	Apply Laplace transform and Fourier transform in signal analysis.
C204.3	Analyze linear time invariant continuous time systems using Laplace and Fourier Transforms.
C204.4	Analyze discrete time signals using Z transform and DTFT.
C204.5	Interpret the linear time invariant discrete time systems using Z transform and DTFT.
Course Code & Title : EC8392 Digital Electronics Semester & Year of study : III & 2018-19 Course Index : C205	
Couse Index	Course Outcomes
C205.1	Apply the concepts of digital electronics in the present contemporary world.
C205.2	Design and implement various combinational digital circuits using logic gates.
C205.3	Analysis and design synchronous sequential circuits.
C205.4	Design and implement asynchronous sequential circuits.
C205.5	Apply the concepts of memory devices and programmable logic devices in Integrated Circuits
Course Code & Title : EC8391 Control Systems Engineering Semester & Year of study : III & 2018-19 Course Index : C206	
Couse Index	Course Outcomes
C206.1	Perform modeling of control system using various techniques.
C206.2	Obtain the time response and steady state error of control systems.
C206.3	Design various compensators and to analyse the frequency response of the system using various plots.
C206.4	Determine the stability of control systems.
C206.5	Analyse and obtain state space model using state variables.

Course Code & Title : EC8381 Fundamentals of Data Structures in C Laboratory Semester & Year of study : III & 2018-19 Course Index : C207	
Couse Index	Course Outcomes
C207.1	Develop C programs for simple applications making use of basic constructs.
C207.2	Apply basic data structures for a given problem using C.
C207.3	Implement linear and non-linear data structures using C.
C207.4	Implement functions and recursive functions in C.
C207.5	Choose appropriate searching, sorting and hashing algorithm for an application and implement it in a modularized way.
Course Code & Title : EC8361 Analog and Digital Circuits Laboratory Semester & Year of study : III & 2018-19 Course Index : C208	
Couse Index	Course Outcomes
C208.1	Plot the frequency response of CE, CB, CC & CS amplifiers.
C208.2	Measure CMRR in differential amplifier.
C208.3	Analyze the limitation in bandwidth of single stage and multistage amplifiers.
C208.4	Simulate the amplifiers using SPICE tool.
C208.5	Design and implement combinational and sequential logic circuits.
Course Code & Title : HS8381 Interpersonal Skills/Listening & Speaking Semester & Year of study : III & 2018-19 Course Index : C209	
Couse Index	Course Outcomes
C209.1	Speak effectively on various academic topics and respond to questions.
C209.2	Converse effectively with the use of conversation starters and discourse markers.
C209.3	Listen and respond to various academic dialogues and discussions.
C209.4	Participate confidently and appropriately in informal and formal conversations and group discussions.
C209.5	Use a range of presentation tools like PPT, Videos, and Charts etc. to make an engaging presentation.

Course Code & Title : MA8451 Probability and Random Processes Semester & Year of study : IV & 2018-19 Course Index : C210	
Couse Index	Course Outcomes
C210.1	Use the fundamental knowledge of the concepts of probability and standard distributions which can describe real life phenomenon.
C210.2	Apply the basic concepts of one and two dimensional random variables in engineering applications.
C210.3	Apply the concept random processes in engineering disciplines.
C210.4	Apply the concept of correlation and spectral densities.
C210.5	Analyse the response of random inputs to linear time invariant systems.
Course Code & Title : EC8452 Electronic Circuits II Semester & Year of study : IV & 2018-19 Course Index : C211	
Couse Index	Course Outcomes
C211.1	Analyze the concepts of Feedback Amplifiers in various applications
C211.2	Design different types of Oscillator at different frequencies.
C211.3	Analyze the performance of Tuned amplifiers
C211.4	Design Pulse circuits and Multivibrators
C211.5	Apply the various design techniques to analyze Power Amplifiers and DC convertors
Course Code & Title : EC8491 Communication Theory Semester & Year of study : IV & 2018-19 Course Index : C212	
Couse Index	Course Outcomes
C212.1	Analyze the various modulation techniques used for communication.
C212.2	Elaborate the angle modulation and demodulation techniques.
C212.3	Apply the concepts of Random Process.
C212.4	Analyze the noise performance of AM and FM systems.
C212 .5	Design applications using the various types of sampling and quantization

Course Code & Title : EC8451 Electromagnetic Fields Semester & Year of study : IV & 2018-19 Course Index : C213	
Couse Index	Course Outcomes
C213.1	Apply the basic mathematical concepts of vector analysis.
C213.2	Describe the laws associated to static electric field and the properties of conductors and dielectrics.
C213.3	Analyze the field potentials due to static magnetic fields and explain how materials affect electric and magnetic fields.
C213.4	Analyze the relation between the fields under time varying situations and apply Maxwell's equations to electric and magnetic fields.
C213.5	Explain electromagnetic wave propagation in lossy and in lossless media.
Course Code & Title : EC8453 Linear Integrated Circuits Semester & Year of study : IV & 2018-19 Course Index : C214	
Couse Index	Course Outcomes
C214.1	Design linear and non-linear applications of op-amps.
C214.2	Design applications using Analog multipliers and PLL.
C214.3	Design ADC and DAC using op-amps.
C214.4	Design waveform Generators using op-amps.
C214.5	Analyze special function ICs.
Course Code & Title : GE8291 Environmental Science and Engineering Semester & Year of study : IV & 2018-19 Course Index : C215	
Couse Index	Course Outcomes
C215.1	Summarize the importance of environment, biodiversity, ecosystem and how to solve environmental related problems.
C215.2	Describe the causes, effect and control measures of air pollution, water pollution, soil pollution, noise pollution, radioactive pollution and thermal pollution with their relevant case studies.
C215.3	Discuss the various renewable and non-renewable resources and energy conservation processes.
C215.4	Explain the social issues and solutions for sustainable environment with relevant Acts and case studies.
C215.5	Review the impact of human population in the environment and its remedial measures.

Course Code & Title : EC8461 Circuits Design and Simulation Laboratory Semester & Year of study : IV & 2018-19 Course Index : C216	
Couse Index	Course Outcomes
C216.1	Differentiate feedback amplifiers with oscillators
C216.2	Calculate the frequency response & the output impedance for various types of feedback amplifiers
C216.3	Design different types of RC, LC oscillators and tuned amplifiers.
C216.4	Analyze the various types of wave-shaping circuits and multivibrators.
C216.5	Simulate oscillators, tuned amplifiers and power amplifiers using SPICE tool
Course Code & Title : EC8462 Linear Integrated Circuits Laboratory Semester & Year of study : IV & 2018-19 Course Index : C217	
Couse Index	Course Outcomes
C217.1	Analyze the basics of linear integrated circuits and available ICs.
C217.2	Design the oscillators, amplifiers and filters using operational amplifiers.
C217.3	Analyze and implement the frequency multiplier using PLL.
C217.4	Design DC power supply using ICs.
C217.5	Analyze the performance of filters, Multivibrators, A/D converters and analog multiplier using SPICE.
III Year	
Course Code & Title : EC8501 Digital Communication Semester & Year of study : V & 2019-20 Course Index : C301	
Couse Index	Course Outcomes
C301.1	Design applications using the various source coding techniques.
C301.2	Interpret the various waveform coding schemes and their representation.
C301.3	Analyze the various baseband transmission schemes.
C301.4	Develop applications using the various band pass signalling schemes.

C301.5	Apply the basic concepts of channel coding techniques.
Course Code & Title : EC8553 Discrete Time Signal Processing Semester & Year of study : V & 2019-20 Course Index : C302	
Couse Index	Course Outcomes
C302.1	Apply DFT for the analysis of digital signals & systems.
C302.2	Design Infinite Impulse response (IIR) digital filters.
C302.3	Design Finite Impulse response (FIR) digital filters.
C302.4	Analyze the finite Word length effects in digital filters.
C302.5	Explain the functionalities and architecture of DSP processors.
Course Code & Title : EC8552 Computer Architecture and Organization Semester & Year of study : V & 2019-20 Course Index : C303	
Couse Index	Course Outcomes
C303.1	Analyze the performance of the computer system and understand the different instructions formats in MIPS architecture.
C303.2	Illustrate the internals of arithmetic and logic units for fixed point and floating point operations.
C303.3	Describe the purposes of data path and control path, pipeline for execution of instructions and its hazards.
C303.4	Explain the various memory organizations with its performances, internal communications methodologies for I/O devices.
C303.5	Interpret the various parallel processing architectures, principles and their challenges.
Course Code & Title : EC8551 Communication Networks Semester & Year of study : V & 2019-20 Course Index : C304	
Couse Index	Course Outcomes
C304.1	Describe the concepts of the network fundamentals and different layers.
C304.2	Identify the components required to build different types of networks and internetworking protocols.
C304.3	Apply the concept of various protocols in routing and multicasting.

C304.4	Explain the flow of information from one node to another in the networks.
C304.5	Analyze the operations of various application layer protocols such as WWW, HTTP, and DNS.
Course Code & Title : EC6504 Medical Electronics Semester & Year of study : V & 2019-20 Course Index : C305	
Couse Index	Course Outcomes
C305.1	Explain about the physiological parameters and recording methods.
C305.2	Analyze the bio-chemicals and various physiological information.
C305.3	Describe various assist devices used in hospitals.
C305.4	Explain the equipment used for physical medicine and the various recently developed diagnostic and therapeutic techniques.
C305.5	Apply the concepts of medical Instrumentation in recent technology (Radio pill, Telemedicine, Endomicroscopy unit).
Course Code & Title : OCE551 Air Pollution and Control Engineering Semester & Year of study : V & 2019-20 Course Index : C306	
Couse Index	Course Outcomes
C306.1	Describe the nature and characteristics of air pollutants with its impact on human, vegetation, animals and properties.
C306.2	Review the meteorological effects of air pollution
C306.3	Explain the design stacks and particulate air pollution control devices to meet applicable standards.
C306.4	Identify equipment's to control gaseous contaminants.
C306.5	Summarize the sources, types and control measures of indoor air pollutants and noise pollution.
Course Code & Title : OIT551 Database Management Systems Semester & Year of study : V & 2019-20 Course Index : C307	
Couse Index	Course Outcomes
C307.1	Interpret relational data model, evolve conceptual model of a given problem, its mapping to relational model.
C307.2	Query the relational database and write simple and complex queries
C307.3	Develop and write programs with database connectivity.
C307.4	Analyze and design a database system and perform normalization.

C307.5	Identify the concepts of database security and information retrieval systems.
Course Code & Title : OME551 Energy Conservation & Management Semester & Year of study : V & 2019-20 Course Index : C308	
Couse Index	Course Outcomes
C308.1	Explain energy auditing methodology
C308.2	Illustrate possible economic measures in electrical systems.
C308.3	Discuss possible economic measures in thermal systems.
C308.4	Describe various energy saving opportunities in major utilities.
C308.5	Analyze the payback period for energy conservation opportunities.
Course Code & Title : OCH551 Industrial Nanotechnology Semester & Year of study : V & 2019-20 Course Index : C309	
Couse Index	Course Outcomes
C309.1	Describe the principle of nanotechnology in electric and electronic devices.
C309.2	Illustrate the therapeutic applications of bio nanotechnology.
C309.3	Explain the concepts of nanotechnology in chemical industries.
C309.4	Summarize the applications of nanotechnology based agriculture and food industries.
C309.5	Discuss the contemporary industrial applications of nanotechnology in textiles and cosmetics.
Course Code & Title : OMF551 Product Design and Development Semester & Year of study : V & 2019-20 Course Index : C310	
Couse Index	Course Outcomes
C310.1	Describe the concepts of product development processes and Organizations
C310.2	Outline the principles of opportunity identification, creating, identifying and screening ideas for new products
C310.3	Explain the implications of product architecture on product change, product variety, component standardization, product performance, manufacturing cost and project management
C310.4	Exhibit the techniques of industrial design of a product through prototyping technology
C310.5	Discuss various techniques used to reduce manufacturing cost and decrease the environmental impacts associated with products
Course Code & Title : ORO551 Renewable Energy Resources Semester & Year of study : V & 2019-20 Course Index : C311	

Couse Index	Course Outcomes
C311.1	Explain the physics of solar radiation
C311.2	Discuss the classification of solar energy collectors and methodologies of storing solar energy.
C311.3	Illustrate the concepts of solar energy utilization in a useful way and applications of solar energy.
C311.4	Describe the concepts in wind energy and biomass with its economic aspects.
C311.5	Analyze in capturing and applying other forms of energy sources like geothermal, Wave, Tidal, OTEC, mini-hydel energies
Course Code & Title : EC8562 Digital Signal Processing Laboratory Semester & Year of study : V & 2019-20 Course Index : C312	
Couse Index	Course Outcomes
C312.1	Generate various signals using MATLAB and DSP processor
C312.2	Implement Linear and circular convolution programs and Frequency Analysis using DFT in MATLAB
C312.3	Implement Auto correlation and Cross Correlation using MATLAB
C312.4	Design FIR and IIR Filters using MATLAB and DSP Processor
C312.5	Analyze the architecture of a DSP Processor and to implement Up-sampling and Down-sampling operation in DSP Processor
Course Code & Title : EC8561 Communication Systems Laboratory Semester & Year of study : V & 2019-20 Course Index : C313	
Couse Index	Course Outcomes
C313.1	Analyze the effects of sampling and TDM
C313.2	Design AM & FM modulation and demodulation
C313.3	Implement Pulse Code Modulation and Delta Modulation
C313.4	Implement the signal constellations of Digital Modulation schemes
C313.5	Implement various Error control coding schemes
Course Code & Title : EC8563 Communication Networks Laboratory Semester & Year of study : V & 2019-20 Course Index : C314	
Couse Index	Course Outcomes

C314.1	Perform client-server communication between two desktop computers using Socket Programming.
C314.2	Implement the different protocols.
C314.3	Simulate various network topologies like Star, Bus and Ring.
C314.4	Implement and compare the different routing algorithms.
C314.5	Simulate the algorithms with the help of Network Simulator tool.
Course Code & Title : EC8691 Microprocessors and Microcontrollers Semester & Year of study : V & 2019-20 Course Index : C315	
Couse Index	Course Outcomes
C315.1	Describe the architecture of microprocessor 8086 and execute programs based on 8086 microprocessor.
C315.2	Explain about design aspects of I/O and Memory Interfacing circuits.
C315.3	Interface 8086 microprocessors with supporting chips.
C315.4	Describe the architecture of microcontroller 8051.
C315.5	Implement 8051 microcontroller based systems.
Course Code & Title : EC8095 VLSI Design Semester & Year of study : V & 2019-20 Course Index : C316	
Couse Index	Course Outcomes
C316.1	Realize the concepts of digital building blocks using MOS transistor.
C316.2	Design combinational MOS circuits and power strategies.
C316.3	Design and construct Sequential Circuits and Timing systems.
C316.4	Design arithmetic building blocks and memory subsystems.
C316.5	Apply and implement FPGA design flow and testing.
Course Code & Title : EC8652 Wireless Communications Semester & Year of study : VI & 2019-20 Course Index : C317	
Couse Index	Course Outcomes

C317.1	Elaborate the characteristics of a wireless channel and evolve the system design specifications
C317.2	Apply the various cellular concepts like frequency reuse, channel assignments, handoff strategies etc., in mobile communication.
C317.3	Analyze the performance of various digital signalling schemes of fading channels.
C317.4	Apply the multipath mitigation techniques based on the application.
C317.5	Implement the concept of transmit/receive diversity in MIMO systems.
Course Code & Title : MG8591 Principles of Management Semester & Year of study : VI & 2019-20 Course Index : C318	
Couse Index	Course Outcomes
C318.1	Discuss the evolution of management, functions and roles of managers.
C318.2	Explain the different types of planning process and tools used for planning.
C318.3	Elaborate different organization structures and functions of human resources manager.
C318.4	Interpret the concepts in motivation techniques, leadership and communication processes
C318.5	Describe the control techniques and the role of technology in management.
Course Code & Title : EC8651 Transmission Lines and RF Systems Semester & Year of study : VI & 2019-20 Course Index : C319	
Couse Index	Course Outcomes
C319.1	Analyze the various types of transmission lines and the losses associated.
C319.2	Analyze different parameters and constraints in high frequency transmission of information.
C319.3	Analyze impedance matching by stubs using smith charts.
C319.4	Analyze the characteristics of TE and TM waves in Guided systems.
C319.5	Design a RF transceiver system for wireless communication.
Course Code & Title : CS8792 Cryptography and Network Security Semester & Year of study : VI & 2019-20 Course Index : C320	

Couse Index	Course Outcomes
C320.1	Describe the fundamentals of networks security, security architecture, threats and vulnerabilities.
C320.2	Apply the different cryptographic operations of symmetric cryptographic algorithms.
C320.3	Explain the different cryptographic operations of public key cryptography.
C320.4	Apply the various Authentication schemes to simulate different applications.
C320.5	Discuss the various Security practices and System security standards.
Course Code & Title : EC8001 MEMS and NEMS Semester & Year of study : VI & 2019-20 Course Index : C321	
Couse Index	Course Outcomes
C321.1	Find solution to Micro/Nano electromechanical systems including their applications and advantages.
C321.2	Recognize the materials in micro fabrication and describe the fabrication processes including surface micromachining, bulk micromachining and LIGA.
C321.3	Analyze the key performance aspects of electromechanical transducers including sensors.
C321.4	Explain the Various electromechanical actuators.
C321.5	Describe the techniques of quantum mechanics and Nano systems.
Course Code & Title : EC8004 Wireless Networks Semester & Year of study : VI & 2019-20 Course Index : C322	
Couse Index	Course Outcomes
C322.1	Analyze different standards used for Wireless LAN.
C322.2	Apply the concepts of Network Layer protocols used for Mobile Applications.
C322.3	Apply the suitable network depending on the availability and requirement.
C322.4	Implement different type of applications for smart phones and mobile devices with latest network strategies.
C322.5	Converse with the latest 3G/4G and WiMAX networks and its architecture.

Course Code & Title : EC8681 Microprocessors and Microcontrollers Laboratory Semester & Year of study : VI & 2019-20 Course Index : C323	
Couse Index	Course Outcomes
C322.1	Write ALP programmes for arithmetic operation, logical operations and data movement using 8086 microprocessor instructions.
C323.2	Implement ALP programmes for code conversion, decimal arithmetic and matrix operations using 8086 instructions.
C323.3	Generate result for floating point operations, string manipulations, sorting, Searching, Password checking, Print RAM size, System Date, Counters and Time Delay using 8086 microprocessor and MASM software.
C323.4	Design 8086/8051 based systems using peripherals and interfaces.
C323.5	Calculate outputs for arithmetic operation, logical operation, square of a number and cube of a number using 8051 microcontroller/MASAM software.
Course Code & Title : EC8661 VLSI Design Laboratory Semester & Year of study : VI & 2019-20 Course Index : C324	
Couse Index	Course Outcomes
C324.1	Write HDL code for basic as well as advanced digital integrated circuits.
C324.2	Synthesize, Place and Route the digital circuits.
C324.3	Import the logic modules in to FPGA boards.
C324.4	Design, Simulate and Extract the layouts of the digital circuits using EDA platforms.
C324.5	Design and Simulate the analog circuits using EDA platforms.
Course Code & Title : EC8611 Technical Seminar Semester & Year of study : VI & 2019-20 Course Index : C325	
Couse Index	Course Outcomes
C325.1	Explain the significance of learning recent advancement in electrical and electronics engineering discipline.
C325.2	Review and prepare the State-of-art technologies in the present technological developments.

C325.3	Organize the presentation using the concepts of ordering and determining the central, main and supporting ideas.
C325.4	Present any topic in any recent advancement with good communicative skill in front of peers and faculty members.
C325.5	Perform well in placement recruitment drive with good technical skills and communication skills.
Course Code & Title : HS8581 Professional Communication Semester & Year of study : VI & 2019-20 Course Index : C326	
Couse Index	Course Outcomes
C326.1	Exhibit soft skills and awareness of different cultures in varied contexts.
C326.2	Make effective presentations.
C326.3	Participate confidently in Group Discussions.
C326.4	Attend job interviews and be successful in them.
C326.5	Set short-term and long-term career goals.
IV Year	
Course Code & Title : EC8701 Antennas and Microwave Engineering Semester & Year of study : VII & 2020-21 Course Index : C401	
Couse Index	Course Outcomes
C401.1	Analyze the basic antenna parameters and link power budget.
C401.2	Describe the design and radiation mechanism of various types of antennas.
C401.3	Explain about the various kinds of antenna arrays.
C401.4	Explain the basic concept of various microwave devices
C401.5	Design a microwave system for the given application.
Course Code & Title : EC8751 Optical Communication Semester & Year of study : VII & 2020-21 Course Index : C402	
Couse Index	Course Outcomes
C402.1	Apply the fundamental concept of optical fiber modes and their configurations.
C402.2	Analyze the various signal degradation factors associated with optical fiber.

C402.3	Explain the Various optical sources and optical detectors and their use in the optical communication system.
C402.4	Apply the techniques required to measure the optical fiber systems based on the applications.
C402.5	Analyze the Digital Transmission and its associated parameters on system performance
Course Code & Title : EC8791 Embedded and Real Time Systems Semester & Year of study : VII & 2020-21 Course Index : C403	
Couse Index	Course Outcomes
C403.1	Explain the fundamental concepts of designing and the computing required for Embedded Systems.
C403.2	Describe the architecture and programming of ARM processor.
C403.3	Apply the programming concepts in embedded system.
C403.4	Analyze the techniques required for creating Real Time Embedded Systems.
C403.5	Apply the concepts of scheduling in Real Time Operating System and creating the model for Real Time applications.
Course Code & Title : EC8702 Adhoc and Wireless Sensor Networks Semester & Year of study : VII & 2020-21 Course Index : C404	
Couse Index	Course Outcomes
C404.1	Explain the Basics of Adhoc networks and Wireless Sensor Networks
C404.2	Apply suitable routing algorithm based on network and user requirement
C404.3	Identify appropriate physical and MAC Layer protocols
C404.4	Describe the transport layer and security issues possible in wireless sensor networks
C404.5	Apply sensor network platforms and tools for various applications.
Course Code & Title : EC8071 Cognitive Radio Semester & Year of study : VII & 2020-21 Course Index : C405	
Couse Index	Course Outcomes
C405.1	Analyze the design principles on software defined radio and cognitive radio.
C405.2	Analyze basic architecture and standards for cognitive radio.

C405.3	Develop the ability to design and implement algorithms for cognitive radio spectrum sensing and dynamic spectrum access.
C405.4	Analyze the MAC and Network layer design for cognitive radio.
C405.5	Apply the knowledge of advanced features of cognitive radio for real world applications.
Course Code & Title : CS8082 Machine Learning Techniques Semester & Year of study : VII & 2020-21 Course Index : C406	
Couse Index	Course Outcomes
C406.1	Differentiate between supervised, unsupervised, semi-supervised machine learning approaches.
C406.2	Apply specific supervised or unsupervised machine learning algorithm for a particular problem.
C406.3	Analyze and suggest the appropriate machine learning approach for the various types of problem.
C406.4	Design and make modifications to existing machine learning algorithms to suit an individual application.
C406.5	Implement various case studies on the advanced machine learning algorithms.
Course Code & Title : OCH752 Energy Technology Semester & Year of study : VII & 2020-21 Course Index : C407	
Couse Index	Course Outcomes
C407.1	Discuss energy scenario, energy calculations, causes of energy scarcity and its solution.
C407.2	Explain different types of power plants, their functions and flow lines.
C407.3	Examine basic properties of various renewable sources of energy and technologies for their utilization.
C407.4	Elaborate biomass energy resources and biomass energy conversion processes
C407.5	Describe the energy conservation measures in process industries.
Course Code & Title : OEN751 Green Building Design Semester & Year of study : VII & 2020-21 Course Index : C408	
Couse Index	Course Outcomes
C408.1	Understand the Environmental Implications of buildings.
C408.2	Enumerate the Embodied Energy of building materials and alternate sustainable concepts.
C408.3	Illustrate the concepts of thermal comfort in buildings and heat transfer

	characteristics of building materials and technologies.
C408.4	Identify the concepts of utility of Solar Energy in buildings.
C408.5	Explain about the Green Composites and Water Utilization in buildings.
Course Code & Title : OCS752 Introduction to C Programming Semester & Year of study : VII & 2020-21 Course Index : C409	
Couse Index	Course Outcomes
C409.1	Implement C programs using basic programming constructs.
C409.2	Develop simple applications using arrays.
C409.3	Design and implement Programs using strings.
C409.4	Develop simple applications using functions.
C409.5	Implement C program using structures.
Course Code & Title : OAN751 Low Cost Automation Semester & Year of study : VII & 2020-21 Course Index : C410	
Couse Index	Course Outcomes
C410.1	Describe the importance of automation in assembly lines.
C410.2	Summarize the features and functions of components to build hydraulics circuits.
C410.3	Explain the working of different pneumatic circuits and systems.
C410.4	Apply the fundamental concepts of sensors with processor/controller interfacing in various applications.
C410.5	Develop some assembly automation.
Course Code & Title : OIE751 Robotics Semester & Year of study : VII & 2020-21 Course Index : C411	
Couse Index	Course Outcomes
C411.1	Explain the concepts of industrial robots, classification, specifications and coordinate systems. Also summarize the need and application of robots in different sectors.
C411.2	Illustrate the different types of robot drive systems as well as robot end effectors.
C411.3	Apply the different sensors and image processing techniques in robotics to

	improve the ability of robots.
C411.4	Develop robotic programs for different tasks and familiarize with the kinematics motions of robot.
C411.5	Examine the implementation of robots in various industrial sectors and interpolate the economic analysis of robots.
Course Code & Title : OML751 Testing Of Materials Semester & Year of study : VII & 2020-21 Course Index : C412	
Couse Index	Course Outcomes
C412.1	Learn about different type of materials, classification and Testing Standards.
C412.2	Analyze engineering components using various mechanical testing procedure.
C412.3	Explain about Principles, Techniques, Advantages and Limitations, Applications of Non-Destructive Testing.
C412.4	Exploit Principles, Types, Advantages and Limitations, Applications of materials characterization process.
C412.5	Ability to use the different technique and know its applications and limitations
Course Code & Title : OCE751 Environmental and Social Impact Assessment Semester & Year of study : VII & 2020-21 Course Index : C413	
Couse Index	Course Outcomes
C413.1	Gain an insight about necessity to study the impacts of development on environment.
C413.2	Carry out scoping and screening of developmental projects for environmental and social assessments and explain different methodologies for environmental impact prediction and assessment.
C413.3	Plan environmental impact assessments, environmental management plans and evaluate environmental impact assessment reports.
C413.4	Carry out economic valuation of environmental impacts.
C413.5	Conduct case studies on different types of projects pertaining EIA.
Course Code & Title : EC8711 Embedded Laboratory Semester & Year of study : VII & 2020-21 Course Index : C414	
Couse Index	Course Outcomes
C414.1	Write programs in ARM for a specific Application.

C414.2	Interface memory with ARM processor and write a program related to memory operations.
C414.3	Interface A/D and D/A convertors with ARM system.
C414.4	Analyze the performance of interrupt.
C414.5	Write programs for interfacing keyboard, display, motor and sensor.
Course Code & Title : EC8761 Advanced Communication Laboratory Semester & Year of study : VII & 2020-21 Course Index : C415	
Couse Index	Course Outcomes
C415.1	Analyze the performance of simple optical link by measurement of losses
C415.2	Analyze the mode characteristics of fiber, eye pattern and the impact on BER
C415.3	Estimate the wireless channel characteristics and analyze the performance of wireless communication system
C415.4	Understand the intricacies in microwave system design and analyze the characteristics of Directional Couplers, Isolators, Circulators
C415.5	Understand the characteristics of Gunn diode and Microwave IC filter
Course Code & Title : GE8076 Professional Ethics in Engineering Semester & Year of study : VIII & 2020-21 Course Index : C416	
Couse Index	Course Outcomes
C416.1	Apply ethics, morals and human values in society
C416.2	Explain about engineering ethics
C416.3	Describe the responsibilities of engineers as experimenters
C416.4	Analyze the safety, risks, risk benefit analysis and rights of an engineer
C416.5	Discuss the importance of the global issues, moral leadership and code of conduct
Course Code & Title : EC8094 Satellite Communication Semester & Year of study : VIII & 2020-21 Course Index : C417	
Couse Index	Course Outcomes
C417.1	Determine the azimuth and elevation angles and visibility of a geostationary satellite from an earth station.

C417.2	Explain the concept of signal propagation of space segment components and create link budgets for an uplink and a downlink
C417.3	Analyze the effect of rain attenuation in a satellite link and the availability of the link based on geographic location of the earth terminals.
C417.4	Design satellite communication system to carry voice, video or data signals using analog or digital modulation.
C417.5	Analyze the various types of satellite services according to its applications.
Course Code & Title : EC8811 Project Work Semester & Year of study : VIII & 2020-21 Course Index : C418	
Couse Index	Course Outcomes
C418.1	Apply the fundamental knowledge and skills, which are acquired within the technical area, to a given problem
C418.2	- Identify and summarize an appropriate list of literature review, analyze previous researchers' work and relate them to the project - Within given constraints, even with limited information, the students will be able to independently analyze and discuss complex inquiries/problems and handle larger problems on the advanced level within the technical area.
C418.3	- Design engineering solutions to complex problems in a systematic approach. - Identify and apply appropriate parameters, assumptions and design criteria in consideration of health and safety (example: the use of codes of practice), ethics, economics, environment, sustainability
C418.4	Apply research and conduct experiments, as well as to analyze and interpret data that yield the results and answer important applicable research questions.
C418.5	Utilize technology tools for communication, collaboration, information management, and decision support.
C418.6	Demonstrate the knowledge, skills and attitudes of a professional engineer.
C418.7	Interact with team members in a professional manner, respecting differences, to ensure a collaborative project environment.
C418.8	Demonstrate a strong working knowledge of ethics and professional responsibility.
C418.9	Document and present one's own work, for a given target group, with strict requirements on structure, format, and language usage.
C418.10	Present the project outlining the approach and expected results using good oral and written presentation skills.
C418.11	Demonstrate effective organizational leadership and change skills for managing projects and project teams.
C418.12	Recognize the need for life-long learning by undergoing the project work.